

Sean Kerr - 3 phase stand-alone micro-grid 40 kW system for 100 kWh daily load and 2.5 days of autonomy - DESIGN SUMMARY

String Array Implementation:

The array design centered around supporting a daily load of 100kWh during the critical month (December) with an anticipated 3 sun hour exposure (direct irradiance at 1000W/m²) daily and a conservative system conversion efficiency of 80% (**32 kW instantaneous array output and 96kWh at 3 sun hours and 80% efficiency**). The modules chosen were Sunpower 335W ($V_{mp} = 36.8W$, $V_{oc} = 44.1V$, $I_{mp} = 9.11A$, $I_{sc} = 9.76A$). To remain within the MPPT input voltage/current max ratings on the inverters, the strings were designed to consist of 15 modules connected in series. The arrangement for the entire array (divided into array 1 and array 2 according to their respective inverters) consists of 120 optimized modules, producing 40 kW at full irradiance, divided into 8 strings, connected into 8 MPPT inputs across 2 inverters. The array is ground mounted and has a default summer angle of 30 degrees (latitude -5) and a steeper winter angle of 45. The strings also have 15A fuses in front of their individual disconnects, and can be shutdown at the module level via the optimizers.

Inverter Selection:

The inverters (Wattsonic 25K-100-3P) were chosen to exceed the desired max array power output at 1/2 40 kW (20kW/inverter). The inverter system consists of 2, 25W hybrid Wattsonic 3 phase inverters (for a total inverter capacity of 50kW). Each inverter has 4 MPPT inputs being fed by a 15 module, optimized string at 760V_{oc} and 552V_{mp} (with the max inverter MPPT inputs MAX V_{mp} rated at 850V). The max AC output current is 42A. The AC disconnects also have OCPD according to the conductor ampacities.

BESS Design:

Built around the Bluetti Energy Pty Ltd 7*B700 51.6 kWh, operating at the rated 500V (with a rated discharge capacity of 50 kWh; but Im considering a more conservative 85% DoD resulting in ~44 kWh or 88 aH capacity per battery). Each inverter has 3 Bluetti batteries in parallel, resulting in a total system charge capacity of 528 aH or 264 kWh at 85% DoD, supporting >2.5 days of full-load autonomy at zero array power output. The battery disconnect is controlled at the inverter battery connection point.

The system is metered at the output to the micro-grid it supplies and has the primary PV system 4P disconnect and OCPD before the meter at the main control panel.

SEE THE CALCULATIONS PAGE FOR ARRAY/INVERTER/BESS/CONDUCTOR AMPACITY/OCPD SIZING

Coldest Day Max VOC Sample Calculation (based on SunPower 335W module temperature coefficient and -40 degree coldest day):

$$\text{Coldest Day Max } V_{oc} = 15 \cdot 44.1V[1 + ((25^{\circ}C - (-40^{\circ}C)) \cdot 0.0029)]$$

$$\text{Coldest Day Max } V_{oc} = 786.2V$$